



EXAM PAPERS PRACTICE

Boost your performance and confidence with these topic-based exam questions

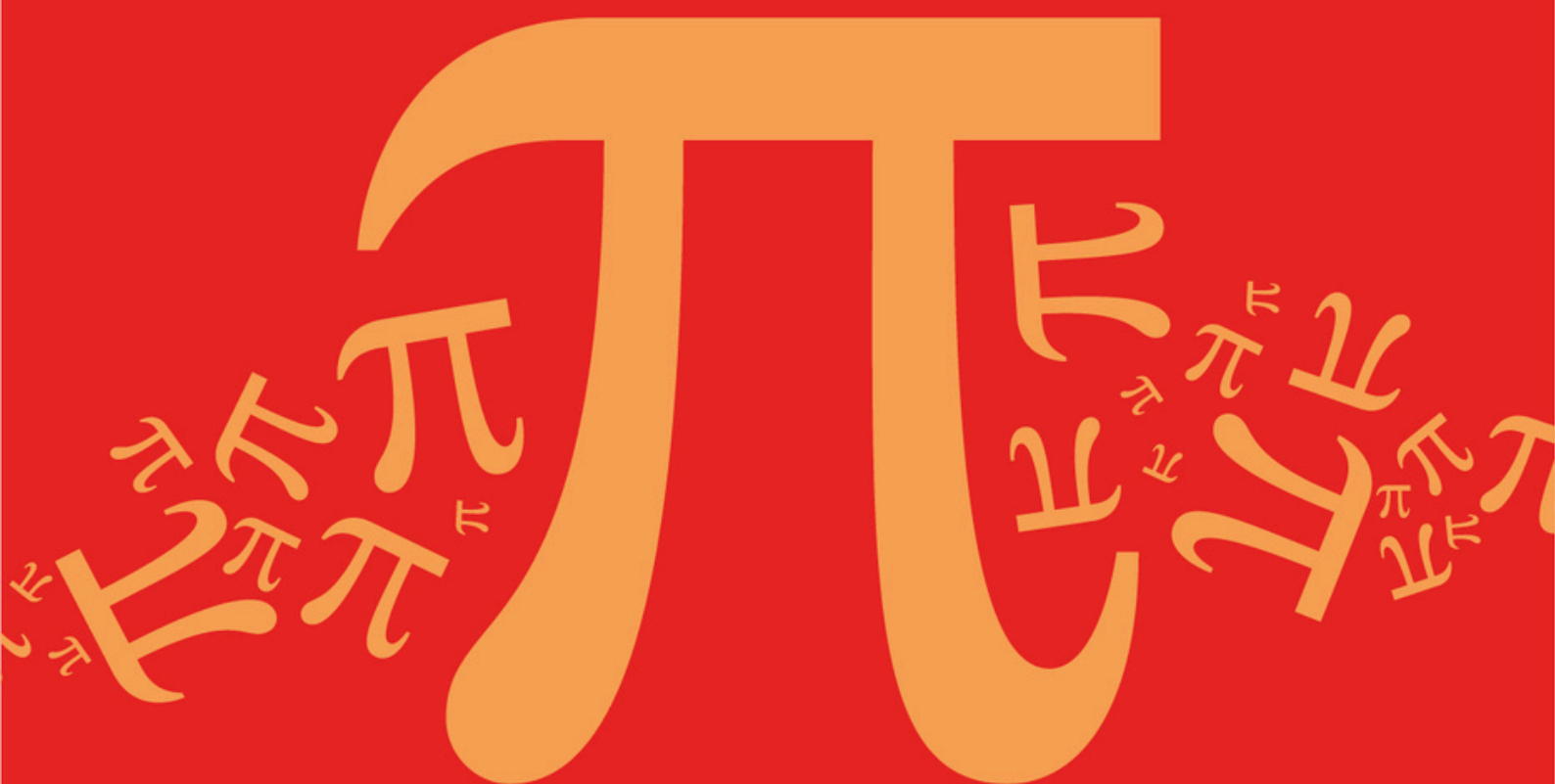
Practice questions created by actual examiners and assessment experts

Detailed mark scheme

Suitable for all boards

Designed to test your ability and thoroughly prepare you

AQA A Level Further Mathematics (7367)



Topic

B: Complex Numbers

Sub topic

B5: Modulus, Argument, and Cartesian Forms

Topic Questions

AQA A Level Further Mathematics (7367)

Question Paper

Topic

B: Complex Numbers

Sub topic

B5: Modulus, Argument, and Cartesian Forms

EXAM PAPERS PRACTICE

© 2024 Exams Papers Practice. All Rights Reserved

**To be used by all students and teachers for
AQA A Level Further Mathematics (7367)
for exams from 2025 onwards.**

Students and teachers of other boards may also find this useful

Name : _____

Date : _____

Time : _____

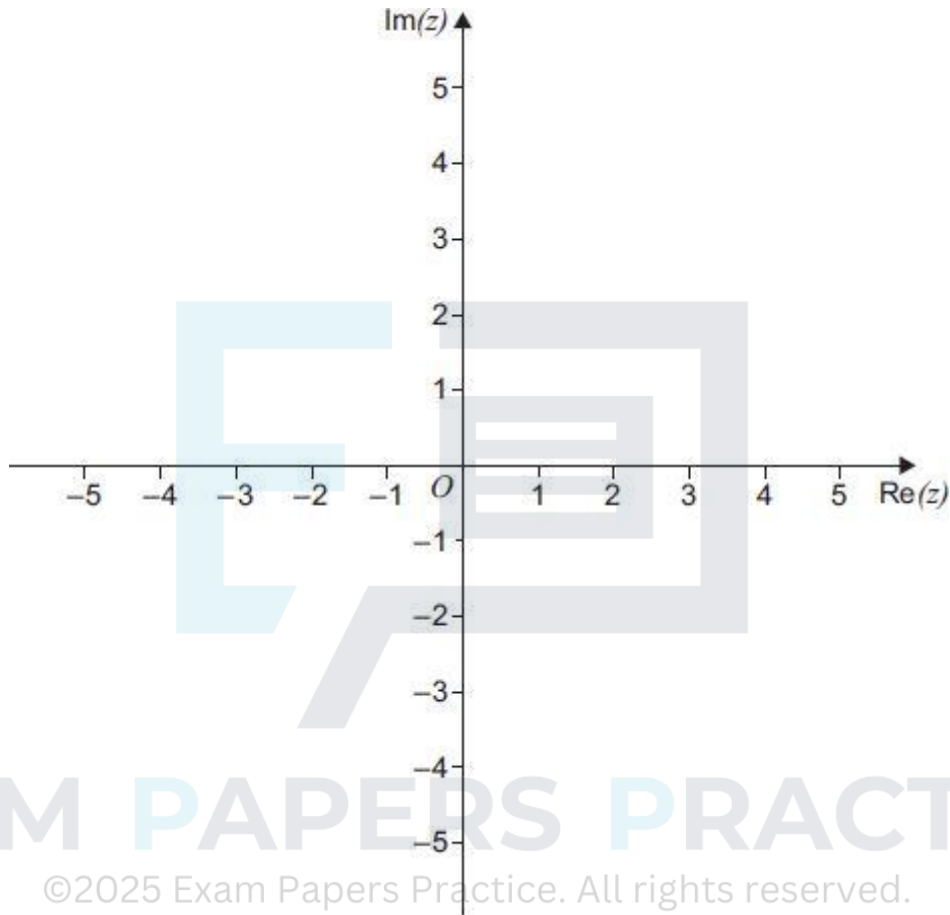
Total Marks Available : _____

Total Marks Received : _____

Q1.

- (a) Sketch, on the Argand diagram below, the locus of points satisfying the equation

$$|z - 3| = 2$$



(1)

- (b) There is a unique complex number w that satisfies both

$$|w - 3| = 2 \text{ and } \arg(w + 1) = \alpha$$

where α is a constant such that $0 < \alpha < \pi$

- (i) Find the value of α .

(2)

- (ii) Express w in the form $r(\cos \theta + i \sin \theta)$.

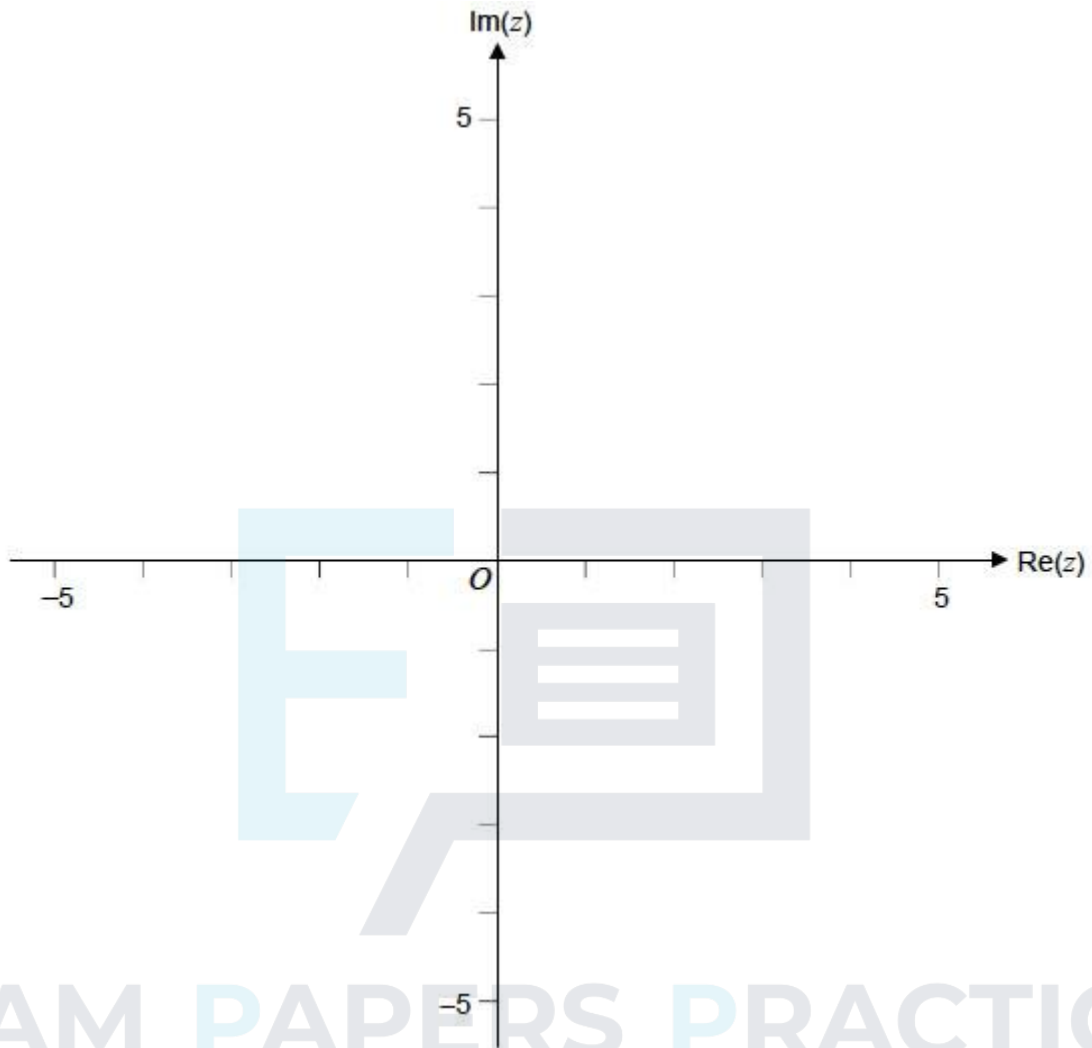
Give each of r and θ to two significant figures.

(4)

(Total 7 marks)

Q2.

- (a) Sketch on the Argand diagram below, the locus of points satisfying the equation
 $|z - 2| = 2$



©2025 Exam Papers Practice. All rights reserved.

(2)

- (b) Given that $|z - 2| = 2$ and $\arg(z - 2) = -\frac{\pi}{3}$, express z in the form $a + bi$, where a and b are real numbers.

(3)

(Total 5 marks)

Q3.

Find the smallest value of θ for which

$$(\cos \theta + i \sin \theta)^5 = \frac{1}{\sqrt{2}}(1 - i) \quad \{\theta \in \mathbb{R} : \theta > 0\}$$

(Total 4 marks)

Q4.

Two loci, L_1 and L_2 , in an Argand diagram are given by

$$L_1 : |z + 6 - 5i| = 4\sqrt{2}$$

$$L_2 : \arg(z + i) = \frac{3\pi}{4}$$

The point P represents the complex number $-2 + i$.

- (a) Verify that the point P is a point of intersection of L_1 and L_2 .

(2)

- (b) Sketch L_1 and L_2 on one Argand diagram.

(6)

- (c) The point Q is also a point of intersection of L_1 and L_2 . Find the complex number that is represented by Q .

(2)

(Total 10 marks)

Q5.

- (a) Express $-4 + 4\sqrt{3}i$ in the form $re^{i\theta}$, where $r > 0$ and $-\pi < \theta \leq \pi$.

(3)

- (b) (i) Solve the equation $z^3 = -4 + 4\sqrt{3}i$, giving your answers in the form $re^{i\theta}$, where $r > 0$ and $-\pi < \theta \leq \pi$.

(4)

- (ii) The roots of the equation $z^3 = -4 + 4\sqrt{3}i$ are represented by the points P , Q and R on an Argand diagram.

Find the area of the triangle PQR , giving your answer in the form $k\sqrt{3}$, where k is an integer.

©2025 Exam Papers Practice. All rights reserved.

(3)

- (c) By considering the roots of the equation $z^3 = -4 + 4\sqrt{3}i$, show that

$$\cos \frac{2\pi}{9} + \cos \frac{4\pi}{9} + \cos \frac{8\pi}{9} = 0$$

(4)

(Total 14 marks)

Q6.

- (a) Sketch on an Argand diagram the locus of points satisfying the equation

$$|z - 6i| = 3$$

(3)

- (b) It is given that z satisfies the equation $|z - 6i| = 3$.

- (i) Write down the greatest possible value of $|z|$.

(1)

- (ii) Find the greatest possible value of $\arg z$, giving your answer in the form $p\pi$,

where $-1 < p \leq 1$.

(3)

(Total 7 marks)

Q7.

(a) Draw on the same Argand diagram:

(i) the locus of points for which

$$|z - 2 - 5i| = 5$$

(3)

(ii) the locus of points for which

$$\arg(z + 2i) = \frac{\pi}{4}$$

(3)

(b) Indicate on your diagram the set of points satisfying both

$$|z - 2 - 5i| \leq 5$$

and

$$\arg(z + 2i) = \frac{\pi}{4}$$

(2)

(Total 8 marks)

Q8.

A circle C and a half-line L have equations

$$|z - 2\sqrt{3} - i| = 4$$

and

$$\arg(z + i) = \frac{\pi}{6}$$

respectively.

(a) Show that:

(i) the circle C passes through the point where $z = -i$;

(2)

(ii) the half-line L passes through the centre of C .

(3)

(b) On one Argand diagram, sketch C and L .

(4)

(c) Shade on your sketch the set of points satisfying both

$$|z - 2\sqrt{3} - i| \leq 4$$

and $0 \leq \arg(z + i) \leq \frac{\pi}{6}$

(2)

(Total 11 marks)

Q9.

- (a) A circle C in the Argand diagram has equation

$$|z + 5 - i| = \sqrt{2}$$

Write down its radius and the complex number representing its centre.

(2)

- (b) A half-line L in the Argand diagram has equation

$$\arg(z + 2i) = \frac{3\pi}{4}$$

Show that $z_1 = -4 + 2i$ lies on L .

(2)

- (c) (i) Show that $z_1 = -4 + 2i$ also lies on C .

(1)

- (ii) Hence show that L touches C .

(3)

- (iii) Sketch L and C on one Argand diagram.

(2)

- (d) The complex number z_2 lies on C and is such that $\arg(z_2 + 2i)$ has as great a value as possible.

Indicate the position of z_2 on your sketch.

(2)

(Total 12 marks)

Q10.

The complex numbers z_1 and z_2 are given by

$$z_1 = \frac{1+i}{1-i} \quad \text{and} \quad z_2 = \frac{1}{2} + \frac{\sqrt{3}}{2}i$$

- (a) Show that $z_1 = i$.

(2)

- (b) Show that $|z_1| = |z_2|$.

- (2)
- (c) Express both z_1 and z_2 in the form $re^{i\theta}$, where $r > 0$ and $-\pi < \theta \leq \pi$. (3)
- (d) Draw an Argand diagram to show the points representing z_1 , z_2 and $z_1 + z_2$. (2)
- (e) Use your Argand diagram to show that

$$\tan \frac{5}{12} \pi = 2 + \sqrt{3}$$

(3)
(Total 12 marks)

Q11.

It is given that $z = e^{i\theta}$.

- (a) (i) Show that

$$z + \frac{1}{z} = 2 \cos \theta$$

(2)

- (ii) Find a similar expression for

$$z^2 + \frac{1}{z^2}$$

(2)

- (iii) Hence show that

$$z^2 - z + 2 - \frac{1}{z} + \frac{1}{z^2} = 4 \cos^2 \theta - 2 \cos \theta$$

(3)

- (b) Hence solve the quartic equation

$$z^4 - z^3 + 2z^2 - z + 1 = 0$$

giving the roots in the form $a + ib$.

(5)
(Total 12 marks)